

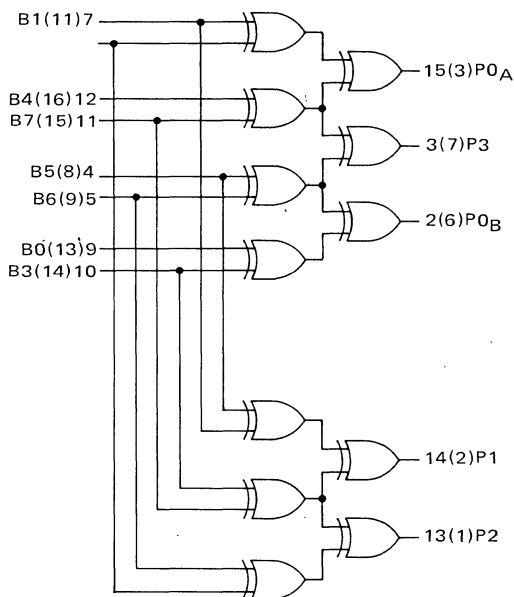
MC10163/MC10563 MC10193/MC10593

ERROR DETECTION - CORRECTION CIRCUITS

The MC10163/MC10563 and the MC10193/MC10593 are error detection and correction circuits. They are building blocks designed for use with memory systems. They offer economy in the design of error detection/correction subsystems for main-frame and add-on memory systems. For example, using eight MC10163's together with eight 12-bit parity checkers (MC10160), single-bit error detection/correction

and double-bit error detection can be done on a word of 64-bit length. Only eight check bits (B0-B7) need be added to the word. A useful feature of this building block is that the MC10193/MC10593 option generates the parity of all inputs to the block. Thus, if the MC10193 is applied in a byte sequence, individual byte parity is automatically available.

MC10163/MC10563 LOGIC DIAGRAM



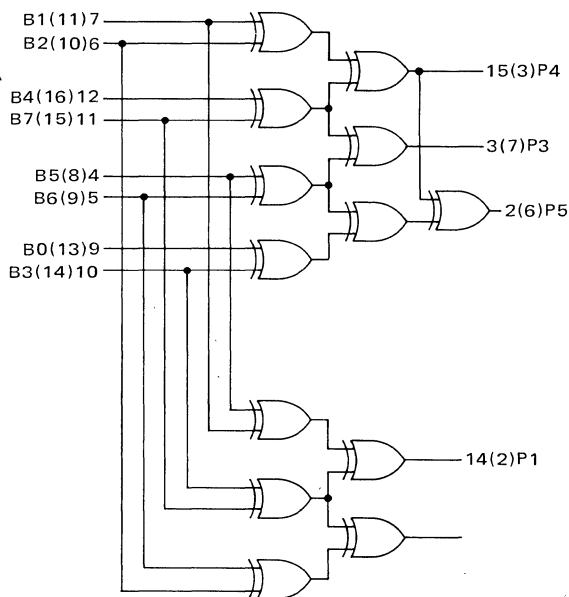
IBM CODE

$P0_A = B1, B2, B4, B7$
 $P0_B = B0, B3, B5, B6$
 $P1 = B1, B3, B5, B7$
 $P2 = B2, B3, B6, B7$
 $P3 = B4, B5, B6, B7$
 $P_D = 520 \text{ mW typ/pkg (No Load)}$
 $t_{pd} = 5.0 \text{ ns typ}$

$V_{CC1} = \text{Pin } 1(5)$
 $V_{CC2} = \text{Pin } 16(4)$
 $V_{EE} = \text{Pin } 8(12)$

Numbers at ends of terminals denote pin numbers for L and P packages.
 Numbers in parenthesis denote pin numbers for F package.

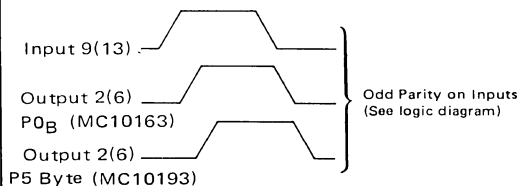
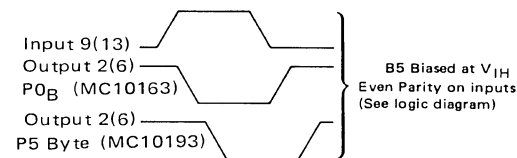
MC10193/MC10593 LOGIC DIAGRAM



MOTOROLA CODE

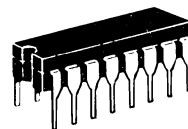
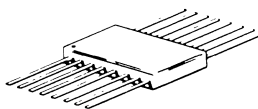
$P1 = B1, B3, B5, B7$
 $P2 = B2, B3, B6, B7$
 $P3 = B4, B5, B6, B7$
 $P4 = B1, B2, B4, B7$
 $P5 = \text{Byte } (B0, 1, 2, 3, 4, 5, 6, 7)$
 $t_{pd} = 7.5 \text{ ns typ (to P5)}$
 $= 5.0 \text{ ns typ (to P1-P4)}$

SWITCHING TIME WAVEFORMS @ 25°C



P SUFFIX
PLASTIC PACKAGE
CASE 648
MC10163 and
MC10193 only

L SUFFIX
CERAMIC PACKAGE
CASE 620



F SUFFIX
CERAMIC PACKAGE
CASE 650
MC10563 and
MC10593 only

Characteristic	Symbol	-55°C		-30°C		+25°C		+85°C		+125°C		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Power Supply Drain Current	I_E	—	137	—	137	—	125	—	137	—	137	mAdc
Input Current	I_{inH}	—	375	—	350	—	220	—	220	—	220	μ Adc
Pins 4,6,10		—	450	—	425	—	265	—	265	—	265	
Pins 5,7,9,11,12												
Switching Times												ns
Propagation Delay	t_{pd}											
MC10163/MC10563		1.3	7.0	1.3	6.8	1.5	6.5	1.5	7.1	1.5	7.5	
MC10193/MC10593 B to P1-P4		1.3	7.1	1.3	6.8	1.5	6.5	1.5	7.1	1.5	11	
B to P5		1.8	9.1	1.8	8.9	2.0	8.5	2.0	9.2	2.0	10	
Rise Time, Fall Time (20% to 80%)	t_r, t_f											ns
MC10163/MC10563		1.1	4.4	1.1	4.2	1.1	3.9	1.1	4.4	1.1	4.5	
MC10193/MC10593		1.1	4.3	1.1	4.2	1.1	3.9	1.1	4.4	1.1	4.6	

-55°C and +125°C test values apply to MC105xx devices only.

MC10163/MC10563 APPLICATIONS INFORMATION

The MC10163/MC10563 is a building block for generating the modified Hamming single-error-correction, double-error-detection (SEC-DED) code used in the IBM370/145 memory. While the MC10163 can also be used for generating other patterns, it is optimized for generating the pattern shown in the H matrix of Figure 1.

When writing into a memory, the MC10163 is used to generate the eight check bits (C0-C32, CT) which are stored with the 64 data bits (B0-B63). These check bits are generated by taking the parity of all data bits marked with an X in the appropriate row of the H matrix. (C0, C1, C32, CT, are even parity; C2, C4, C8, C17, are odd parity.) To generate these check bits with the building blocks, eight MC10163's and eight MC10160 parity checkers are used. One MC10163 is connected to each byte of data and the outputs of these building blocks are connected to the eight MC10160 parity checkers, one for each check bit. Figure 2 shows which connections are required (i.e., C0 is the even parity of output P0_A of the MC10163 on the "zero" byte of data, output

P0_B of the "zero" byte, P0_A of the "one" byte, ..., P0_B of the "three byte and data bit 32.)

During the memory read operation, the fetched check bits previously generated (as described) are exclusive-ORed with newly generated C0-C32 to generate syndrome bits S0-S32. Syndrome ST is a special case where ST is the even parity of all eight fetched check bits and all 64 fetched data bits. For determining the type and location of an error:

1. If all syndromes (S0-S32 and ST) are false, there is no error.
2. If ST is true and S0-S32 are false, the CT is in error.
3. If ST is false and one or more of S0-S32 is true, an uncorrectable error has occurred.
4. If ST is true and one or more of S0-S32 is true, simply add the S1-S32 bits to get the binary location of the error (S1 has weight 1, S2 weight 2, S4 weight 4, etc.)

Data bits B0 and B32 are special cases of this location technique: B0 is in error if ST, S0, and S32 are true; B32 is in error if ST, S0, S1, and S32 are true.

FIGURE 1 – 370/145 PATTERN

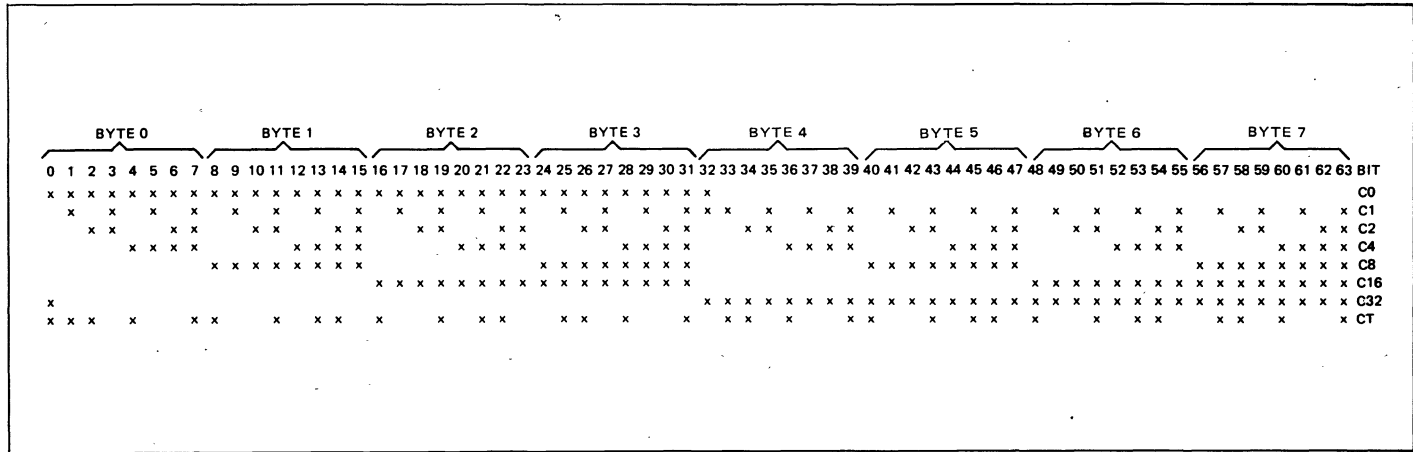


FIGURE 2 – 370/145 PATTERN GENERATION

C0 = P0A0	P0B0	P0A1	P0B1	P0A2	P0B2	P0A3	P0B3	B(32)
C1 = P10	P11	P12	P13	P14	P15	P16	P17	B(32)
C2 = P20	P21	P22	P23	P24	P25	P26	P27	
C4 = P30	P31	P32	P33	P34	P35	P36	P37	
C8 = P0A1	P0B1	P0A3	P0B3	P0A5	P0B5	P0A7	P0B7	
C16 = P0A2	P0A2	P0A3	P0B3	P0A6	P0B6	P0A7	P0B7	
C32 = P0A4	P0B4	P0A5	P0B5	P0A6	P0B6	P0A7	P0B7	B(0)
CT = P0A0	P0B1	P0B2	P0A3	P0A4	P0B5	P0B6	P0A7	B(0)

Where for P_{NM}: N = MC10163 Output
M = Byte Number



Figure 5 gives the error location circuit for the example pattern. The outputs EB0 to EB6 are a one-of-eight-high code giving the byte in error. Outputs EC0 to EC3 give the binary location of the bit in error within the located byte. Since this location process can occur simultaneously with the determination of error type described, the entire error correction sequence (using a toggling fetched data latch) takes less than 20 ns. This is because an error occurrence detector is a simple ORing of S1 to S8. The error locator has simultaneously located the error which is then corrected as through the error was a single (and therefore correctable) error. The parity of syndromes then determines if the error was indeed single, and interrupts the CPU if the error was an uncorrectable (multiple) error. Since uncorrec-

table data is unusable without special handling, the CPU would be interrupted anyway; therefore this automatic correction of any error as if it were single does not create any problems. This fast error correction technique allows

single error correction on a non-interrupt basis with only a 20 ns memory system access time penalty.

These techniques can, of course, be extended to large or smaller data words.

FIGURE 4 – M2 PATTERN BUILDING BLOCK

$S1 = P10 \ P11 \ P12 \ P13 \ P54 \ P55 \ P56 \ B(64)$
 $S2 = P20 \ P21 \ P22 \ P23 \ P54 \ P55 \ P57 \ B(65)$
 $S3 = P30 \ P31 \ P32 \ P33 \ P54 \ P56 \ P57 \ B(66)$
 $S4 = P40 \ P41 \ P42 \ P43 \ P55 \ P56 \ P57 \ B(67)$
 $S5 = P14 \ P15 \ P16 \ P17 \ P50 \ P51 \ P52 \ B(68)$
 $S6 = P24 \ P25 \ P26 \ P27 \ P50 \ P51 \ P53 \ B(69)$
 $S7 = P34 \ P35 \ P36 \ P37 \ P50 \ P52 \ P53 \ B(70)$
 $S8 = P44 \ P45 \ P46 \ P47 \ P51 \ P52 \ P53 \ B(71)$

Where for P_{NM} : N = MC10193 Output
M = Byte Number

FIGURE 5 – M2 PATTERN CORRECTION MATRIX

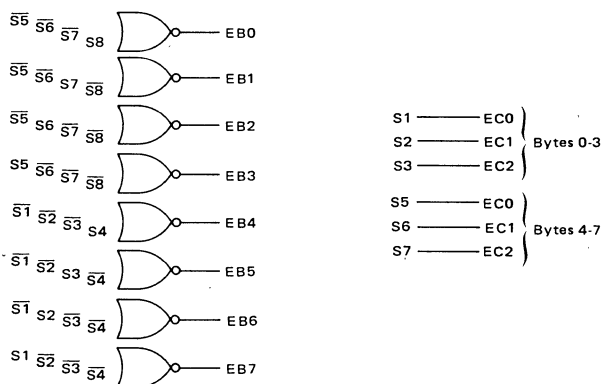


Figure 1 is a block diagram of the MC10193 and MC10160. The diagram shows eight MC10193 blocks (labeled 0-7) and eight MC10160 blocks (labeled S1-S8). The MC10193 blocks are connected to the MC10160 blocks via a series of OR gates. The inputs to the OR gates are labeled B64 through B71. The outputs of the MC10160 blocks are labeled S1 through S8. The diagram also includes a legend for the check bits: "1" for Check Bits and "0" for Syndromes.